

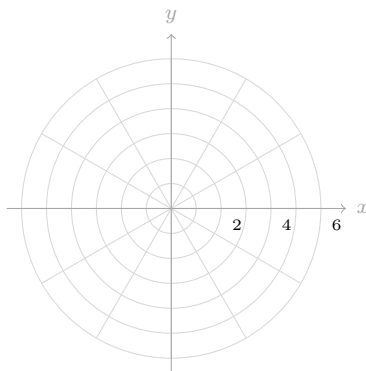
Converting Between Rectangular and Polar Coordinates

Polar to rectangular, rectangular to polar, and choosing the angle that matches the quadrant

Directions. Every problem has a polar grid beside it. Plot the point first — the picture tells you which quadrant the answer must land in, and that is the step a calculator cannot do for you.

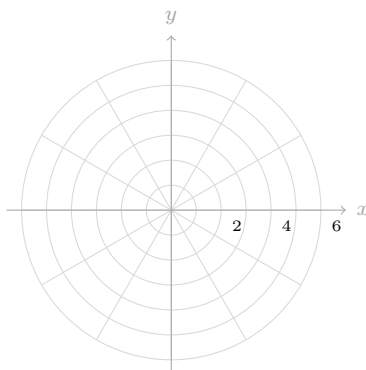
- **Polar $\rightarrow$ rectangular:** $x = r \cos \theta$, $y = r \sin \theta$. Substitute and simplify; leave exact radicals unless a decimal is asked for.
- **Rectangular $\rightarrow$ polar:** $r = \sqrt{x^2 + y^2}$ (a distance, so never negative), and $\tan \theta = \frac{y}{x}$.
- **The angle needs a decision.** $\tan \theta = \frac{y}{x}$ has two solutions on $[0^\circ, 360^\circ)$, half a turn apart, and the arctangent key only ever returns one of them. Plot the point, see the quadrant, and pick the solution that lands there.

- 1.** The polar point $(r, \theta) = (6, \frac{\pi}{3})$. Plot it, then convert it to rectangular form. Give x and y exactly.



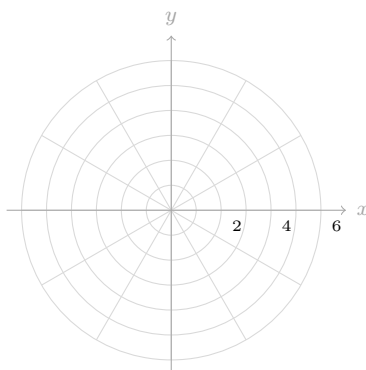
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- 2.** The rectangular point $(3, 4)$. Plot it, then find r exactly and θ in degrees to two decimal places. Which quadrant is the point in, and does the arctangent key need any correction here?



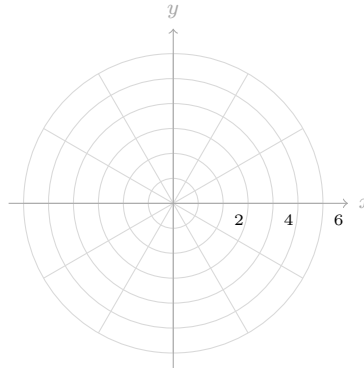
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- 3.** The polar point $(r, \theta) = (4, \frac{3\pi}{4})$. Plot it and convert to rectangular form, exactly. Check the signs of x and y against the quadrant you plotted in.



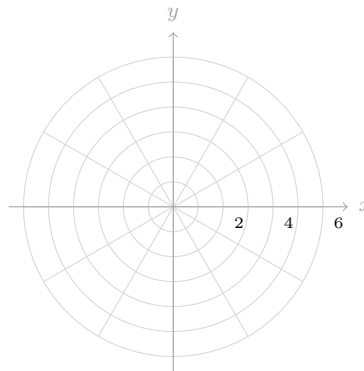
Answer: _____

4. The rectangular point $(-2, 2)$. (a) Find r exactly. (b) Solve $\tan \theta = -1$ on $[0^\circ, 360^\circ)$ — you will get two angles. (c) Say which of the two is the polar angle of this point, and why the other one is wrong.



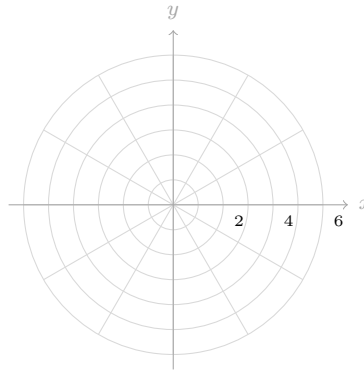
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5. A point has $\tan \theta = \sqrt{3}$ and $r = 4$, and it lies in the third quadrant. (a) Solve $\tan \theta = \sqrt{3}$ on $[0^\circ, 360^\circ)$ to get both candidate angles. (b) Choose the one in the third quadrant and convert the point to rectangular form, exactly.



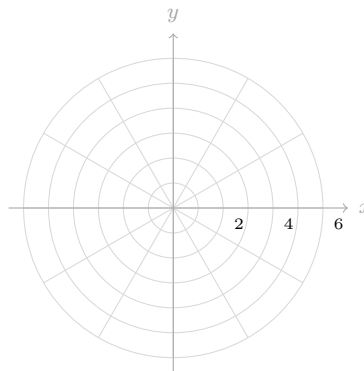
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- 6.** The polar point $(r, \theta) = (6, 210^\circ)$. Plot it and convert to rectangular form, exactly. The angle is in degrees, so convert it before using sin and cos in radian mode.



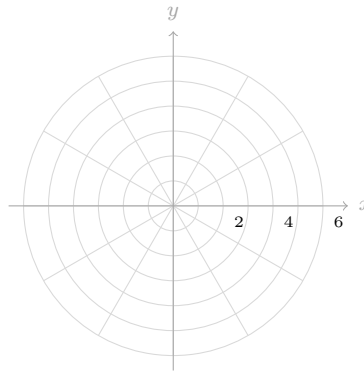
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- 7.** The rectangular point $(-3, -4)$. **(a)** Find r . **(b)** Find θ in degrees to two decimal places. Compare your work with problem 2: the two points use the same right triangle, so what exactly is different about the answer?



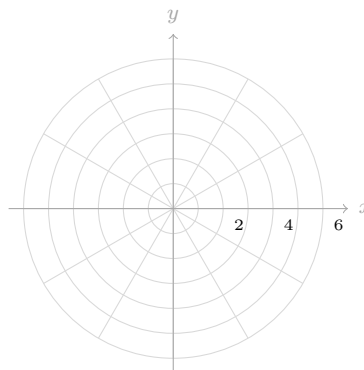
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8. The rectangular point $(-4, 4)$. (a) Find r exactly. (b) Type $\frac{y}{x}$ into your calculator's arctangent key and write down what it returns. (c) Plot the point and give the correct θ in degrees, explaining the adjustment you had to make.



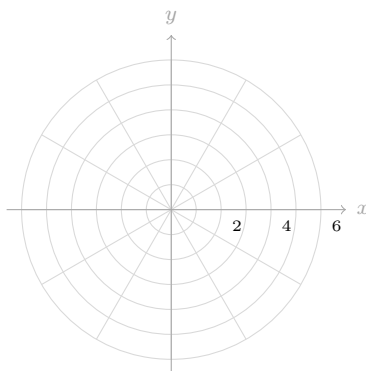
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9. The polar curve $r = 4 \cos \theta$. (a) Find the rectangular coordinates of the point on the curve at $\theta = \frac{\pi}{3}$; give x exactly and y to two decimal places. (b) Multiply the equation through by r , replace r^2 and $r \cos \theta$ using the conversion formulas, and write the curve as a rectangular equation. Name the curve.



Answer: _____

10. The rectangular point $(0, -5)$. **(a)** Find r . **(b)** Since $x = r \cos \theta$ is zero here, solve $\cos \theta = 0$ on $[0^\circ, 360^\circ)$ for the two candidate angles, then use the sign of y to choose. **(c)** Convert your polar answer back to rectangular form as a check, and explain why the $\tan \theta = \frac{y}{x}$ formula cannot be used for a point on the y -axis.



Answer: _____